

Cube Root, Cubic, and Other Polynomial Functions	Unit	CHECKPOINT		
		1	2	3
2A.2 Attributes of functions and their inverses. The student applies mathematical processes to understand that functions have distinct key attributes and understand the relationship between a function and its inverse. 2A.7 Number and algebraic methods. The student applies mathematical processes to simplify and perform operations on expressions and to solve equations. Connected Knowledge and Skills 2A.6				

Process (Tools to Know)	Unit	CHECKPOINT		
		1	2	3
2A.1(A) apply math in everyday situations ⓘ 2A.1(B) use problem-solving models ⓘ <i>connected 2A.1(C)</i>				

Content	Unit	CHECKPOINT		
		1	2	3
Cubic and Cube Root Functions				
2A.2(A) graph the functions $f(x)=\sqrt{x}$, $f(x)=1/x$, $f(x)=x^3$, $f(x)=\sqrt[3]{x}$, $f(x)=b^x$, $f(x)= x $, and $f(x)=\log_b(x)$ where b is 2, 10, and e , and, when applicable, analyze the key attributes such as domain, range, intercepts, symmetries, asymptotic behavior, and maximum and minimum given an interval		Data included in "Absolute Value Functions"		
2A.2(C) describe and analyze the relationship between a function and its inverse (quadratic and square root, logarithmic and exponential), including the restriction(s) on domain, which will restrict its range				
2A.2(B) graph and write the inverse of a function using notation such as $f^{-1}(x)$		Data included in "Quadratic and Square Root Functions"		
2A.2(D) use the composition of two functions, including the necessary restrictions on the domain, to determine if the functions are inverses of each other				
2A.6(A) analyze the effect on the graphs of $f(x) = x^3$ and $f(x) = \sqrt[3]{x}$ when $f(x)$ is replaced by $af(x)$, $f(bx)$, $f(x - c)$, and $f(x) + d$ for specific positive and negative real values of a , b , c , and d				
2A.6(B) solve cube root equations that have real roots				

Polynomial Functions				
2A.7(E) determine linear and quadratic factors of a polynomial expression of degree three and of degree four, including factoring the sum and difference of two cubes and factoring by grouping				
2A.7(B) add, subtract, and multiply polynomials				
2A.7(C) determine the quotient of a polynomial of degree three and of degree four when divided by a polynomial of degree one and of degree two				
2A.7(D) determine the linear factors of a polynomial function of degree three and of degree four using algebraic methods				

Radical Expressions				
2A.7(H) solve equations involving rational exponents				
2A.7(G) rewrite radical expressions that contain variables to equivalent forms				

Process (Ways to Show)	Unit	CHECKPOINT		
		1	2	3
2A.1(E) create representations 2A.1(F) analyze information ⓘ <i>connected 2A.1(D), 2A.1(G)</i>				